

AMY ARIYO

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PROFILE

MSci Earth Sciences graduate specialising in GIS, remote sensing and environmental data analysis. Experienced in spatial-data processing, statistical analysis and map production using QGIS, Python and R, with interests in environmental change, climate risk and land-use analysis.

EDUCATION

University College London (UCL) 2022 – 2026
MSci Earth Sciences, Upper Second-Class Honours (2:1)
Research Project Prize 2026 – Highest-marked MSci dissertation in the Department of Earth Sciences.
Relevant Modules: GIS and Remote Sensing, Artificial Intelligence for Earth Observation, Statistics for Geoscientists, Environmental Geoscience

GEOSPATIAL AND ENVIRONMENTAL PROJECTS

MSci Dissertation – Arctic Sea-Ice Remote Sensing 2025 – 2026

- Co-located MISR optical roughness and SARAL/AltiKa radar-altimetry data across six Arctic dates using projected nearest-neighbour matching.
- Applied waveform classification, geometry filtering and spatial, temporal and along-track sensitivity tests; identified a consistent negative relationship between roughness and radar backscatter.

Urban Land-Cover Classification, Waltham Forest 2024

- Processed five years of Sentinel-2 imagery and produced annual land-cover maps using QGIS, Python and machine-learning classification.
- Compared K-means and Random Forest models; the supervised model achieved 0.81 overall accuracy and 0.72 Cohen's kappa.

UK Rainfall Extremes Analysis 2025

- Analysed multi-decade rainfall data for 12 UK cities to examine spatial, seasonal and year-to-year patterns in extreme events.
- Applied city-specific statistical thresholds and presented findings through maps, heatmaps and trend visualisations, with documented assumptions and limitations.

EXPERIENCE

Nuffield Research Placement, Swansea University 2021
Research Assistant

- Conducted experiments on sodium-ion battery materials, recorded and validated quantitative data and presented findings within an engineering research team.
- Followed laboratory procedures, maintained accurate documentation and learned unfamiliar experimental methods during a six-week placement.

Marks and Spencer 2021 – 2022
Customer Assistant

- Assisted customers, operated tills and supported stock and daily operational tasks while maintaining accuracy in a busy retail environment.
- Resolved queries, managed competing priorities and communicated clearly with customers and colleagues.

SKILLS

GIS & Remote Sensing	QGIS, spatial analysis, raster/vector processing, map production, Sentinel imagery, satellite-data interpretation
Programming & Analysis	Python, pandas, scikit-learn, R, ggplot2, dplyr, statistical analysis and data visualisation
Data Management	Data cleaning, validation, integration, quality assurance, spatial matching and reproducible workflows
Communication & Tools	Technical reporting, presentations, research synthesis, Microsoft Excel, Word, PowerPoint, Google Colab and GitHub